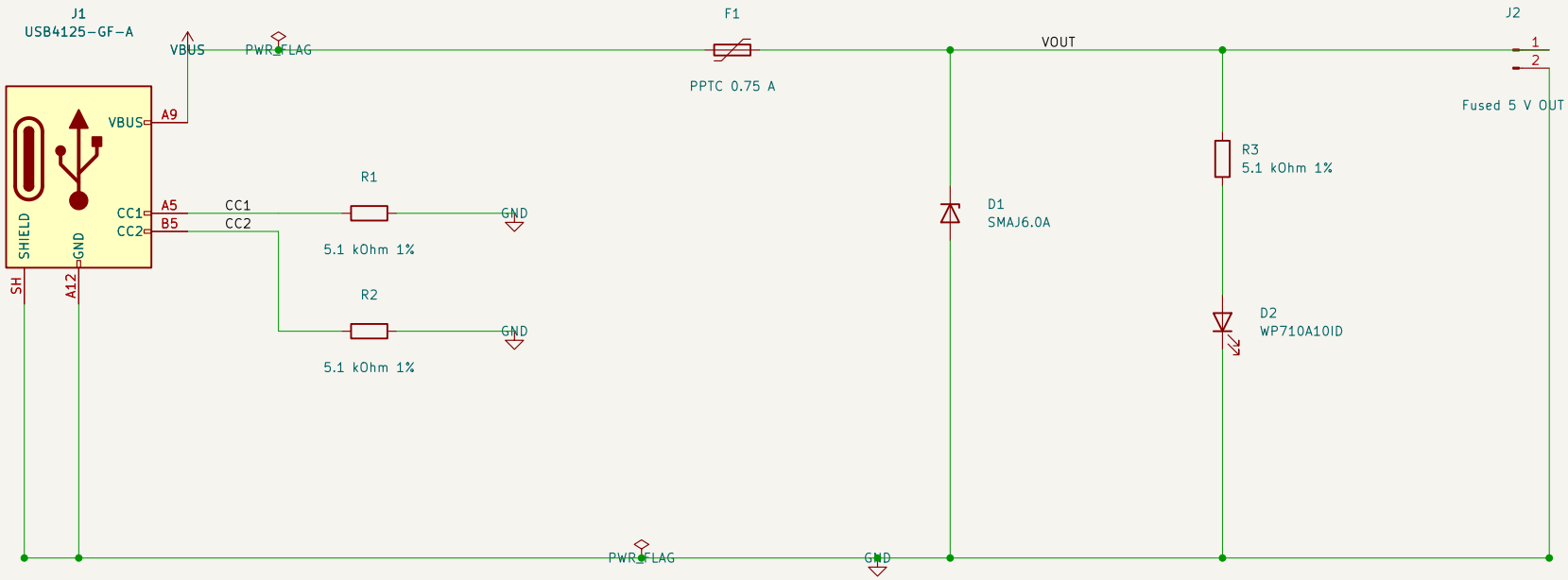
1 USB-C attachment

2 Fuse + transient shunt

3 Output + indicator



H1 M3 NPTH

H2 M3 NPTH

F1: 0.75 A hold / 1.50 A trip at 23 C (not a 500 mA limiter).
D1: 6 V standoff; up to 10.3 V clamp at rated pulse.
TVS is NOT 5 V overvoltage protection for sensitive loads.

R3: I_LED <=1.04 mA at 5.25 V, zero-Vf worst bound.
J2: pin 1 VOUT, pin 2 GND. Never backfeed output.
Shield bonded to GND for this bench example only.

External source: 4.75–5.25 V; total board + load <=500 mA near 23 C.
No PD negotiation or current-advertisement detector.
Two independent 5.1 kOhm Rd resistors; six-contact power-only connector.

DRAFT / UNBUILT — educational bench use only
Sources: GCT USB4125; Bourns MF-MSMF; Vishay SMAJ;
Yageo MFR; Kingbright WP710A10ID; Samtec TSW/HTSW.
Exact PDFs, limits and outstanding fit/bench checks: DESIGN_BASIS.md

No PD negotiation or USB current-advertisement sensing
DRAFT — unbuilt; externally qualified source and load required
PCB Copilot examples

Sheet: /		
File: usb-c-power.kicad_sch		
Title: USB-C 5 V bench power entry		
Size: A4	Date: 2026-10-29	**Rev: B**
KiCad E.D.A. 10.0.6		Id: 1/1